

DENVER INTL, CO, USA

WMO: 725650

Lat: 39.833N

Lon: 104.658W

Elev: 1650

StdP: 83.01

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 03017

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.9	-14.6	-22.0	0.6	-11.2	-19.6	0.8	-6.5	13.3	5.7	11.8	2.7	3.4	230	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	34.9	15.5	33.5	15.4	32.0	15.3	18.2	27.2	17.6	27.0	16.9	26.4	4.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	13.8	20.1	15.0	13.0	19.9	14.1	12.3	19.7	58.3	26.9	55.8	26.8	53.8	26.3	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.6	8.9	-23.5	37.4	3.1	1.1	-25.8	38.2	-27.6	38.9	-29.3	39.5	-31.6	40.3		
(5)			-23.8	19.7	3.0	0.5	-26.0	20.1	-27.8	20.4	-29.5	20.7	-31.7	21.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-0.1	0.4	5.2	8.6	13.7	20.0	23.9	22.7	18.1	10.6	4.6	-0.5		
	DBStd	10.08	6.17	6.42	5.92	5.26	4.90	4.09	2.90	2.86	4.70	5.71	6.14	6.36		
	HDD10.0	1433	314	272	167	86	22	1	0	0	5	62	178	328		
	HDD18.3	3263	571	502	408	292	155	29	2	4	59	243	413	585		
	CDD10.0	1672	2	2	18	44	138	299	431	393	248	81	15	1		
	CDD18.3	459	0	0	0	1	12	78	174	139	52	3	0	0		
	CDH23.3	5609	0	1	6	28	241	1078	1978	1479	697	99	2	0		
	CDH26.7	2520	0	0	0	2	65	487	1011	675	265	15	0	0		
(14) Wind	WSAvg	4.5	4.3	4.5	4.8	5.2	4.6	4.3	4.2	4.2	4.2	4.3	4.2	4.3		
(15) Precipitation	PrecAvg	388	10	12	20	48	58	48	54	49	38	26	12	12		
	PrecMax	548	23	35	68	143	115	102	129	111	190	63	55	53		
	PrecMin	201	0	0	1	4	15	11	8	4	0	4	3	1		
	PrecStd	87	6	8	15	29	31	27	35	28	39	16	10	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.9	24.1	26.2	31.3	36.2	37.0	35.5	33.9	28.4	23.1	17.9		
		MCWB	5.9	6.7	8.5	10.1	13.7	14.6	15.9	15.4	14.9	11.7	8.4	5.9		
	2%	DB	14.7	16.5	21.2	23.8	28.4	33.7	35.3	33.9	31.6	26.2	20.2	14.5		
		MCWB	4.6	5.4	7.5	9.3	12.7	14.7	15.8	15.3	14.0	10.8	7.5	4.5		
	5%	DB	11.8	13.7	18.6	21.6	26.1	31.8	33.9	32.4	29.9	23.8	17.5	11.7		
		MCWB	3.3	4.1	6.6	8.6	12.2	14.5	15.8	15.4	13.4	10.0	6.5	3.3		
	10%	DB	9.0	10.4	15.8	19.0	23.5	29.7	32.4	30.7	27.7	20.7	14.5	8.9		
		MCWB	2.1	2.5	5.6	7.9	11.6	14.5	15.9	15.5	13.2	9.1	5.1	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	7.2	9.4	11.8	15.9	18.6	19.1	18.9	17.5	13.2	9.1	6.4		
		MCDB	16.6	18.5	21.7	21.3	25.9	28.2	28.0	27.0	26.7	24.9	21.1	16.8		
	2%	WB	5.0	5.7	8.1	10.2	14.5	17.2	18.4	18.0	16.1	11.8	7.9	4.9		
		MCDB	13.8	15.8	19.5	20.5	23.6	26.6	27.3	27.0	25.5	22.4	19.3	14.0		
	5%	WB	3.6	4.3	7.0	9.3	13.4	16.3	17.8	17.4	15.1	10.7	6.7	3.5		
		MCDB	11.3	13.0	17.6	19.6	22.6	26.0	27.1	26.8	25.0	21.3	16.8	11.0		
	10%	WB	2.3	2.9	5.8	8.4	12.4	15.5	17.2	16.7	14.3	9.7	5.5	2.1		
		MCDB	8.7	10.0	14.9	18.0	21.4	25.7	26.8	26.0	24.1	19.5	13.9	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	13.3	14.4	14.1	14.0	15.5	15.3	15.0	15.2	14.3	13.9	13.0		
		MCDBR	16.1	17.7	18.1	18.2	17.7	18.4	17.3	17.0	17.6	18.2	17.6	16.3		
		MCWBR	8.7	9.2	8.5	8.0	6.6	5.9	4.8	4.8	5.7	7.4	8.5	9.0		
	5% WB	MCDBR	15.6	16.7	17.2	16.7	15.6	15.8	14.8	14.6	15.2	16.2	16.8	15.6		
		MCWBR	8.8	9.0	8.4	7.7	6.7	6.2	5.0	4.9	5.6	7.1	8.4	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.245	0.265	0.296	0.315	0.324	0.351	0.342	0.306	0.267	0.249	0.239		
	taud		2.623	2.614	2.560	2.483	2.489	2.479	2.441	2.457	2.536	2.633	2.603	2.608		
	Ebn at Noon		968	1001	1003	981	961	949	919	921	942	958	941	939		
	Edh at Noon		66	77	91	105	107	108	112	107	92	75	66	62		
(44) All-Sky Solar Radiation	RadAvg		2.53	3.49	4.73	5.66	6.32	7.22	6.95	6.20	5.29	3.86	2.77	2.22		
	RadStd		0.11	0.20	0.34	0.39	0.50	0.33	0.29	0.18	0.26	0.24	0.15	0.13		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	N/A

Nomenclature: See separate page